

Teoría de perturbaciones

Unidad 3: Teoría de sesgo galáctico

Alejandro Avilés (ICF-UNAM)

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Curso PCF-UNAM/ DCI-UG

Teoría de sesgo

ON THE SPATIAL CORRELATIONS OF ABELL CLUSTERS

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ABSTRACT

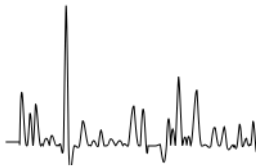
If rich clusters formed where the primordial density enhancement, when averaged over an appropriate scale, was unusually large, then they give a biased measure of the large-scale density correlation function $\xi_{\text{density}}(r)$. The factor A is determined by the probability distribution of the density fluctuations on the cluster mass scale, and if this distribution was Gaussian the correlation function is amplified. The factor A for rich $R \geq 1$ clusters is estimated to be $A \approx 10$, and the predicted trend of A with richness agrees well with that observed. Some implications of these results for the large-scale density correlations are discussed.

Subject headings: cosmology — galaxies: clustering

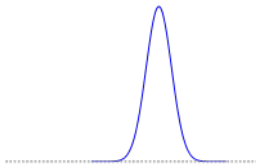
Effective field theory for biased tracers

- We observe biased tracers of the underlying dark matter distribution
- Their properties depend on baryonic and non-linear effects that are outside of the linear regime
- Bias theories are EFT.

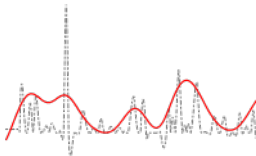
$$(\boldsymbol{x})$$



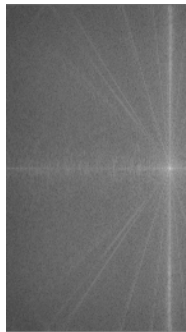
$$W_R(\boldsymbol{x}) = \exp\big[\frac{|\boldsymbol{x}|^2}{2R^2}\big]$$



$${}_R(\boldsymbol{x}) = \int d^3x' W_R(\boldsymbol{x} - \boldsymbol{x}') \, \rho(\boldsymbol{x}')$$

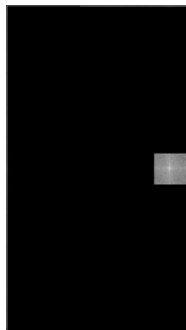


Sebastien Fromenteau experiment



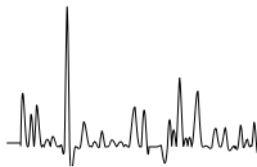
Credit: Sebastien Fromenteau

Sebastien Fromenteau experiment

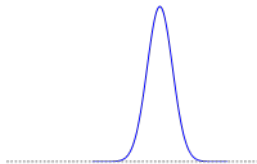


Credit: Sebastien Fromenteau

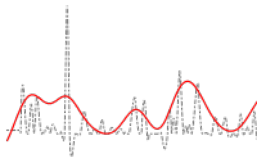
$$\psi(\mathbf{x})$$



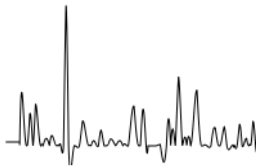
$$W(\mathbf{x})$$



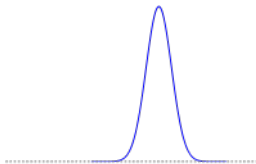
$$\psi(\mathbf{x}) = \int d^3x' W(\mathbf{x} - \mathbf{x}') \psi(\mathbf{x}')$$



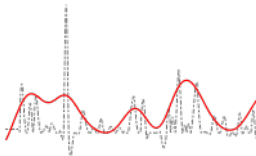
$$(\boldsymbol{x})$$



$$W_R(\boldsymbol{x})$$



$${}_R(\boldsymbol{x}) = \int d^3x' W_R(\boldsymbol{x} - \boldsymbol{x}') \, (\boldsymbol{x}')$$



Bias expansion

Consider tracers X with density number $n_X(\boldsymbol{x})$. The overdensity is

$$\delta_X(\boldsymbol{x}) = \frac{n_X(\boldsymbol{x}) - \bar{n}_X}{\bar{n}_X}, \quad \langle \delta_X(\boldsymbol{x}) \rangle = 0$$

Introduce a bias function

$$1 + \delta_X(\boldsymbol{x}) = F_{\boldsymbol{x}}[\boldsymbol{r}; \boldsymbol{x}]$$

and *local bias* parameters

$$c_n = F^{(n)}(0)$$

Expanding about $\boldsymbol{r} = 0$

$$1 + \delta_X(\boldsymbol{r}) = c_0 + c_1 \delta_R(\boldsymbol{r}) + \frac{1}{2} c_2 \delta_R^2(\boldsymbol{r}) + \frac{1}{6} c_3 \delta_R^3(\boldsymbol{r}) + \dots$$

Up to linear overdensity fields ($\delta_R = \delta_R^{(1)}$), and third order in bias expansion

$$\begin{aligned} 1 + \xi_X(r) &= \langle (1 + \delta_X(\mathbf{r}_1))(1 + \delta_X(\mathbf{r}_2)) \rangle \\ &= \langle (c_0 + c_1 \delta_R(\mathbf{r}_1) + \frac{1}{2} c_2 \delta_R^2(\mathbf{r}_1) + \frac{1}{6} c_3 \delta_R^3(\mathbf{r}_1) + \cdots) \\ &\quad (c_0 + c_1 \delta_R(\mathbf{r}_2) + \frac{1}{2} c_2 \delta_R^2(\mathbf{r}_2) + \frac{1}{6} c_3 \delta_R^3(\mathbf{r}_2) + \cdots) \rangle \\ &= (c_0 + \frac{1}{2} c_2 \sigma_R^2 + \cdots)^2 + (c_1 + \frac{1}{2} c_3 \sigma_R^2 + \cdots)^2 \xi_R(r) + \frac{1}{2} \end{aligned}$$

zero-lag correlator:

$$\sigma_R^2 = \langle \delta_R(\mathbf{r}_1) \delta_R(\mathbf{r}_1) \rangle = \int \frac{d^3k}{(2\pi)^3} |\tilde{W}_R(k)|^2 P_L(k) \approx \int_0^{1/R} \frac{d}{2\pi}$$

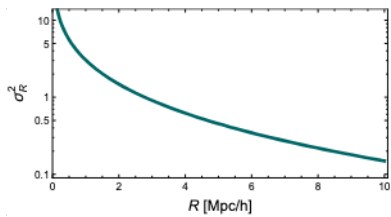
$$\xi_R(r) = \langle \delta_R(\mathbf{r}_2) \delta_R(\mathbf{r}_1) \rangle = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{r}\cdot\mathbf{k}} |\tilde{W}_R(k)|^2 P_L(k)$$

That is

$$\sigma_R^2 = \xi_R(0).$$

$$\sigma_{R\rightarrow 0}^2 \rightarrow \infty$$

Statistics depend on the cutoff scale R even for very large scales $r \gg R$



Renormalization

- Reorganization of bias parameters

$$b_0 = c_0 + \frac{1}{2}c_2\sigma_R^2 + \dots = 1$$

$$b_1 = c_1 + \frac{1}{2}c_3\sigma_R^2 + \dots$$

$$b_2 = c_2 + \dots$$

$$b_3 = c_3 + \dots$$

- Correlation function:

$$\begin{aligned}1 + \xi_X(r) &= (c_0 + \frac{1}{2}c_2\sigma_R^2 + \dots)^2 + (c_1 + \frac{1}{2}c_3\sigma_R^2 + \dots)^2 \\&\quad + \frac{1}{2}(c_2 + \dots)^2\xi_R^2(r) + \dots \\&= 1 + b_1^2\xi_R(r) + \frac{1}{2}b_2^2\xi_R^2(r) + \dots\end{aligned}$$

So far, we have assumed linear order in σ_R .

Poisson shot noise

Define

$$n(\mathbf{x}) = \sum_i \mathcal{D}(\mathbf{x} - \mathbf{x}_i), \quad \bar{n} = \langle n(\mathbf{x}) \rangle, \quad \xi(\mathbf{x}) = \frac{n(\mathbf{x}) - \bar{n}}{\bar{n}}$$

Then

$$n(\mathbf{x})n(\mathbf{y}) = \sum_{i,j} \mathcal{D}(\mathbf{x} - \mathbf{x}_i) \mathcal{D}(\mathbf{y} - \mathbf{x}_j) = \sum_i \mathcal{D}(\mathbf{x} - \mathbf{x}_i) \mathcal{D}(\mathbf{y} - \mathbf{x}_i) + \sum_{i \neq j} \mathcal{D}(\mathbf{x} - \mathbf{x}_i) \mathcal{D}(\mathbf{y} - \mathbf{x}_j)$$

The self-pair term gives

$$\left\langle \sum_i \mathcal{D}(\mathbf{x} - \mathbf{x}_i) \mathcal{D}(\mathbf{y} - \mathbf{x}_i) \right\rangle = \bar{n} \mathcal{D}(\mathbf{x} - \mathbf{y}),$$

while the distinct-pair term gives

$$\left\langle \sum_{i \neq j} \mathcal{D}(\mathbf{x} - \mathbf{x}_i) \mathcal{D}(\mathbf{y} - \mathbf{x}_j) \right\rangle = \bar{n}^2 [1 + \xi(\mathbf{x} - \mathbf{y})].$$

Therefore,

$$\langle n(\boldsymbol{x})n(\boldsymbol{y})\rangle = \bar{n}^2[1 + \xi(\boldsymbol{x} - \boldsymbol{y})] + \bar{n} \, \text{D}(\boldsymbol{x} - \boldsymbol{y}),$$

and hence (defining $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{x}$)

$$\langle n(\boldsymbol{x})n(\boldsymbol{x} + \boldsymbol{r})\rangle = \xi(r) + \frac{1}{\bar{n}} \, \text{D}(\boldsymbol{r}).$$

Fourier transforming ($P(k) = \int d^3r \, e^{i \cdot \boldsymbol{r}} \xi(r)$),

$$P_{\text{tot}}(k) = P(k) + \frac{1}{\bar{n}}.$$

So the constant Poisson shot noise in the power spectrum is the Fourier transform of the Dirac delta function term at $\boldsymbol{r} = 0$.

How deviations from constant shot noise appear

For pure Poisson sampling, the stochastic contribution is only a contact term

$$\xi_{\text{stoch}}(\mathbf{r}) = \frac{1}{\bar{n}} \mathcal{D}(\mathbf{r}),$$

so in Fourier space

$$P_{\text{stoch}}(k) = \frac{1}{\bar{n}}.$$

More generally, if the occupation process is not exactly Poisson,

$$\xi_{\text{stoch}}(\mathbf{r}) = \frac{1}{\bar{n}} \mathcal{D}(\mathbf{r}) + \eta(\mathbf{r}),$$

where $\eta(\mathbf{r})$ is a short-range contribution at finite separation.

Its Fourier transform is

$$P_{\text{stoch}}(k) = \frac{1}{\bar{n}} + \tilde{\eta}(k).$$

Therefore, deviations from constant shot noise arise from non-contact stochastic configuration space.

If $\eta(\boldsymbol{r})$ is isotropic and short-ranged, then at low k

$$P_{\text{stoch}}(k) = P_0 + P_2 k^2 + P_4 k^4 + \dots$$

The constant part P_0 need not equal $1/\bar{n}$:

$$P_0 \neq \frac{1}{\bar{n}}.$$

This can happen because of: **exclusion, correlated halo occupation, luminosity weighting, nonlinear stochastic bias, ...**

Hence:

$$\text{Poisson} \Rightarrow \text{pure } \delta(\boldsymbol{r})$$

while

$$\text{non-Poisson} \Rightarrow \delta(\boldsymbol{r}) + \text{finite-range terms.}$$

- **Exclusion:** Galaxies or halos cannot overlap below a characteristic distance, which suppresses close pairs and often leads to sub-Poisson shot noise.
- **Correlated halo occupation:** The number of galaxies hosted by different halos within the same halo, can be statistically correlated, adding extra variance beyond Poisson sampling.
- **Luminosity or mass weighting:** When objects are weighted by luminosity or another property, the effective shot noise is no longer just $1/\bar{n}$ but depends on the distribution of weights.
- **Nonlinear stochastic bias:** Small-scale galaxy formation physics introduces higher-order terms in the bias expansion, generating both a renormalized constant bias and scale-dependent corrections such as k^2 .

Renormalization

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Clustering of dark matter tracers: Renormalizing the bias par

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Beyond linear δ_R :

$$\delta_X(\mathbf{x}) = c_0 + c_1 \delta_R(\mathbf{x}) + \frac{c_2}{2} \delta_R^2(\mathbf{x}) + \frac{c_3}{3!} \delta_R^3(\mathbf{x}) + \dots$$

$$\delta_X(\mathbf{x}) = \epsilon(\mathbf{x}) + c_1 \delta_R(\mathbf{x}) + \frac{c_2}{2} \int_{12=} \delta_R(\mathbf{x}_1) \delta_R(\mathbf{x}_2) + \frac{c_3}{3!} \int_{123=} \delta_R(\mathbf{x}_1) \delta_R(\mathbf{x}_2) \delta_R(\mathbf{x}_3) + \dots$$

where $\epsilon(\mathbf{x})$ is a stochastic field (uncorrelated with $\delta_R(\mathbf{x})$ at large scales $k \gg k_{\text{NL}}$)

The smoothed matter density field is

$$\begin{aligned} \delta_R(\mathbf{x}) = & \delta_R^{(1)}(\mathbf{x}) + \int_{12=} F_2(\mathbf{x}_1, \mathbf{x}_2) \delta_R^{(1)}(\mathbf{x}_1) \delta_R^{(1)}(\mathbf{x}_2) \\ & + \int_{123=} F_3(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \delta_R^{(1)}(\mathbf{x}_1) \delta_R^{(1)}(\mathbf{x}_2) \delta_R^{(1)}(\mathbf{x}_3) + \dots \end{aligned}$$

Beyond linear δ_R :

$$\delta_X(\mathbf{x}) = c_0 + c_1 \delta_R(\mathbf{x}) + \frac{c_2}{2} \delta_R^2(\mathbf{x}) + \frac{c_3}{3!} \delta_R^3(\mathbf{x}) + \dots$$

$$\delta_X(\mathbf{x}) = \epsilon(\mathbf{x}) + c_1 \delta_R(\mathbf{x}) + \frac{c_2}{2} \int_{12=} \delta_R(\mathbf{x}_1) \delta_R(\mathbf{x}_2) + \frac{c_3}{3!} \int_{123=} \delta_R(\mathbf{x}_1) \delta_R(\mathbf{x}_2) \delta_R(\mathbf{x}_3) + \dots$$

where $\epsilon(\mathbf{x})$ is a stochastic field (uncorrelated with $\delta_R(\mathbf{x})$ at large scales $k \gg k_{\text{NL}}$)

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$$(2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') P_X(\mathbf{k}) = \langle X(\mathbf{k}) X(\mathbf{k}') \rangle$$

Up to third order in bias and matter fluctuations, one obtains (Tarea)

$$P_X(\mathbf{k}) = [c_1^2 + c_1 c_3 \sigma_R^2 + \frac{68}{21} c_1 c_2 \sigma_R^2] P(\mathbf{k}) + \frac{1}{2} c_2^2 \int \frac{d^3 p}{(2\pi)^3} P(p) \\ + 2 c_1 c_2 \int \frac{d^3 p}{(2\pi)^3} P(p) P(|\mathbf{k} - \mathbf{p}|) F_2(\mathbf{p}, -\mathbf{p}) + P_{\text{shot}}$$

with

$$P(\mathbf{k}) = \langle R(\mathbf{k}) R(\mathbf{k}') \rangle', \\ P_{\text{shot}} = \langle \epsilon(\mathbf{k}) \epsilon(\mathbf{k}') \rangle'$$

Shot noise is a constant to leading order. We will derive the shot noise later.

Defining to linear order

$$b_1^2=c_1^2+c_1c_3\sigma_R^2+\frac{68}{21}c_1c_2\sigma_R^2+\cdots$$

$$b_2=c_2+\cdots$$

$$\begin{aligned} P_X(\mathbf{\Lambda}) &= b_1^2 P_{1\text{-loop}}^{\text{SPT}}(\mathbf{\Lambda}) + \frac{1}{2} b_2^2 \int \frac{d^3p}{(2\pi)^3} P_L(p) P_L(|\mathbf{\Lambda} - \mathbf{p}|) \\ &\quad + 2b_1 b_2 \int \frac{d^3p}{(2\pi)^3} P_L(p) P_L(|\mathbf{\Lambda} - \mathbf{p}|) F_2(\mathbf{p}, -\mathbf{p}) \end{aligned}$$

However, the integral

$$\int \frac{d^3p}{(2\pi)^3} P_L(p) P_L(|\mathbf{p}|)$$

is sensitive to large internal momentum, when p . Indeed for a power spectrum which is a good approximation at small scales, it behaves as

$$\int dp p^2 P_L(p) P_L(|\mathbf{p}|) \stackrel{k \ll p}{\rightarrow} \int dp p^{2n+2} \propto$$

which is UV-divergent for $n \geq 3/2$.

When $k \ll p$, it goes to the constant $\int \frac{d^3p}{(2\pi)^3} P_L^2(p)$. One can absorb this constant shot noise:

$$P_{\text{shot}} \rightarrow P_{\text{shot}} + \frac{1}{2} b_2^2 \int \frac{d^3p}{(2\pi)^3} P_L^2(p)$$

and hence

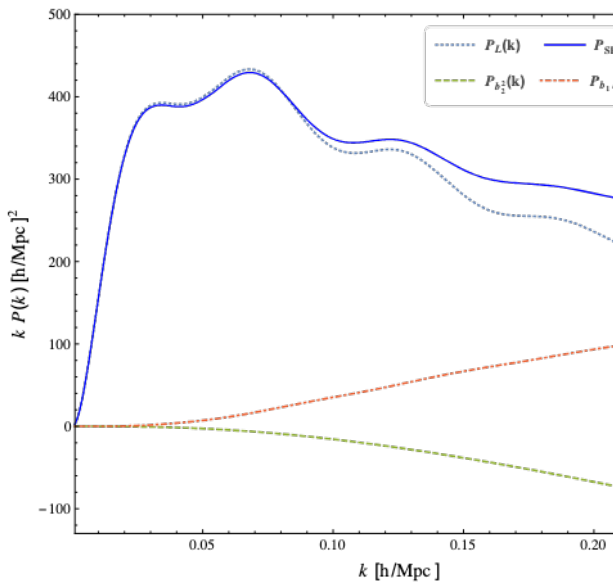
$$\int \frac{d^3p}{(2\pi)^3} P_L(p) P_L(|\mathbf{p}|) \rightarrow \int \frac{d^3p}{(2\pi)^3} P_L(p) P_L(|\mathbf{p}|)$$

which goes to zero at large scale and is UV-divergent only for $n \geq 1/2$ (re UV divergent for $n \geq 1$)

Hence the final expression for the 1-loop SPT power spectrum for tracers with local bias parameters is

$$\begin{aligned}
 P_X(k) = & b_1^2 P_{1\text{-loop}}^{\text{SPT}}(k) + \frac{1}{2} b_2^2 \int \frac{d^3p}{(2\pi)^3} P_L(p) P_L(|\mathbf{k} - \mathbf{p}|) \\
 & + 2b_1 b_2 \int \frac{d^3p}{(2\pi)^3} P_L(p) P_L(|\mathbf{k} - \mathbf{p}|) F_2(\mathbf{p}, -\mathbf{p}) + \dots
 \end{aligned}$$

$$P_X(k) = b_1^2 P_{1\text{-loop}}^{\text{SPT}}(k) + b_2^2 P_{b_2^2}(k) + b_1 b_2 P_{b_1 b_2}(k) + P_{\text{shot}}$$



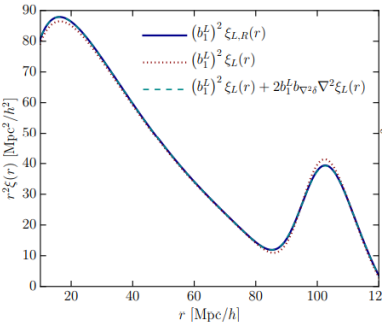
Curvature bias

PHYSICAL REVIEW D **88**, 023515 (2013)

Peak-background split, renormalization, and galaxy clustering

Fabian Schmidt,¹ Donghui Jeong,² and Vincent Desjacques³

• $F_R = F_R + \nabla^2 F_R$



Generalizing bias

Clustering of dark matter tracers: generalizing bias for the coming LSS

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- Beyond Taylor expansion in density fields: $\chi = \sum_n \frac{c_n}{n!} \delta^n$
- Construct bias operators depending on velocity v_i and gravitational Φ
- Equivalence principle implies biasing depends only on $\nabla_i v_j$ and $\nabla_i \nabla_j \Phi$

How the gravitational potential affect the evolution of tracers?

- A homogeneous shift in the gravitational field $\Phi \rightarrow \Phi + \Phi_0$ is not observed. Biasing does not depend on Φ .
- A homogeneous shift in the gravitational force $\nabla^i \Phi \rightarrow \nabla^i \Phi + C^i$ is not observed. Biasing does not depend on $\nabla^i \Phi$.
- Biasing depends on $\nabla_i \nabla_j \Phi$.

Tidal bias

We can construct an *operator*

$$(4\pi G\bar{\rho})^{-1} \nabla_i \nabla_j \Phi.$$

But its trace is $(4\pi G\bar{\rho})^{-1} \nabla^2 \Phi = \dots$, and we already have the bias operator in the expansion of the biasing function $F_{\mathbf{x}}[\dots]$. Hence we subtract the trace to avoid

$$s_{ij} = \frac{1}{4\pi G\bar{\rho}} \left[\nabla_i \nabla_j \Phi - \frac{1}{3} \delta_{ij} \nabla^2 \Phi \right] = \left[\frac{\nabla_i \nabla_j}{\nabla^2} - \frac{1}{3} \delta_{ij} \right]$$

and construct the scalar

$$s^2 \equiv s_{ij} s^{ij}$$

which is the *tidal bias operator*.

$$X = c_1 + \frac{c_2}{2} s^2 + \frac{c_3}{6} s^3 + \dots + b_{s^2} s^2 + \dots$$

Tidal bias

In Fourier space

$$s_{ij}(\mathbf{x}) = \left[\frac{\nabla_i \nabla_j}{\nabla^2} - \frac{1}{3} \delta_{ij} \right] (\mathbf{x}) \rightarrow s_{ij}(\mathbf{k}) = \left[\frac{k_i k_j}{k^2} - \frac{1}{3} \delta_{ij} \right]$$

Then

$$\begin{aligned} s^2(\mathbf{k}) &= \int_{12} s_{ij}(\mathbf{k}_1) s_{ij}(\mathbf{k}_2) \\ &= \int_{12} \left(\frac{k_1^i k_1^j}{k_1^2} - \frac{1}{3} \delta_{ij} \right) \left(\frac{k_2^i k_2^j}{k_2^2} - \frac{1}{3} \delta_{ij} \right) (\mathbf{k}_1) \\ &= \int_{12} S^{(2)}(\mathbf{k}_1, \mathbf{k}_2) (\mathbf{k}_1) (\mathbf{k}_2) \end{aligned}$$

with

$$S^{(2)}(\mathbf{k}_1, \mathbf{k}_2) = \frac{(\mathbf{k}_1 \cdot \mathbf{k}_2)^2}{k_1^2 k_2^2} - \frac{1}{3} = \frac{2}{3} \mathcal{P}_2(\hat{k}_1 \cdot \hat{k}_2),$$

where $\mathcal{P}_2$ is the second order Legendre Polynomial.

Tidal bias is second order (at least).

Bias constructed out of the velocity field

- A velocity field v_i can be removed by a coordinate transformations. The bias depends on $\nabla_i v_i \propto \nabla_i \nabla_j \nabla^2 \theta$.
- But at linear order $\theta = \theta$, and $\nabla_i v_j$ is redundant with $\nabla_i \nabla_j \Phi$ at linear order. To simplify things is to define

$$\eta \equiv \theta \quad (= 0 \text{ at linear order}),$$

and

$$t_{ij} \equiv \left[\frac{\nabla_i \nabla_j}{\nabla^2} - \frac{1}{3} \delta_{ij} \right] \eta.$$

The variables η and t_{ij} are non-zero at second order and beyond.

As with s_{ij} we have to construct scalars:

$$s^2 = s_{ij} s_{ij} \quad (\text{second order})$$

$$st = s_{ij} t_{ij} \quad (\text{third order})$$

$$s^3 = s_{ij} s_{jk} s_{ki} \quad (\text{third order})$$

$$t^2 = t_{ij} t_{ij} \quad (\text{fourth order})$$

In Fourier Space $\eta(\mathbf{k}) = \theta(\mathbf{k}) - \frac{1}{2}(\mathbf{k} \cdot \mathbf{k})$. At second order $\eta^{(2)}(\mathbf{k}) = \theta^{(2)}(\mathbf{k}) - \frac{1}{2}(\mathbf{k} \cdot \mathbf{k})^2$

$$\eta^{(2)}(\mathbf{k}) = \int_{12=} [G_2(\mathbf{k}_1, \mathbf{k}_2) - F_2(\mathbf{k}_1, \mathbf{k}_2)] \theta^{(1)}(\mathbf{k}_1) \theta^{(1)}(\mathbf{k}_2)$$

$$G_2(\mathbf{k}_1, \mathbf{k}_2) = \frac{13}{21} + \frac{4}{7} \left(\frac{(\mathbf{k}_1 \cdot \mathbf{k}_2)^2}{k_1^2 k_2^2} - \frac{1}{3} \right) + \frac{1}{2} \frac{\mathbf{k}_1 \cdot \mathbf{k}_2}{k_1 k_2} \left(\frac{k_1}{k_2} + \frac{k_2}{k_1} \right)$$

$$F_2(\mathbf{k}_1, \mathbf{k}_2) = \frac{17}{21} + \frac{2}{7} \left(\frac{(\mathbf{k}_1 \cdot \mathbf{k}_2)^2}{k_1^2 k_2^2} - \frac{1}{3} \right) + \frac{1}{2} \frac{\mathbf{k}_1 \cdot \mathbf{k}_2}{k_1 k_2} \left(\frac{k_1}{k_2} - \frac{k_2}{k_1} \right)$$

$$G_2 - F_2 = \frac{4}{21} + \frac{2}{7} \left(\frac{(\mathbf{k}_1 \cdot \mathbf{k}_2)^2}{k_1^2 k_2^2} - \frac{1}{3} \right).$$

Then

$$\begin{aligned} \eta^{(2)}(\mathbf{k}) &= \int_{12=} \left[\frac{4}{21} + \frac{2}{7} S_2(\mathbf{k}_1, \mathbf{k}_2) \right] \theta^{(1)}(\mathbf{k}_1) \theta^{(1)}(\mathbf{k}_2) \\ &= \frac{4}{21} \theta^{(2)}(\mathbf{k}) + \frac{2}{7} s^{(2)}(\mathbf{k}) \end{aligned}$$

That is, at second order $\eta = \frac{4}{21} \theta^2 + \frac{2}{7} s^2$

Complete bias expansion

We define

$$\psi(\boldsymbol{x}) = \eta(\boldsymbol{x}) - \frac{2}{7}s^2(\boldsymbol{x}) + \frac{4}{21}s^2(\boldsymbol{x})$$

which is **third order** by construction.

We have the following bias operators

At least 1st order	η, s^2
At least 2nd order	ψ, s^2, s^3
At least 3rd order	ψ, s^2, s^3

Complete bias expansion

From the biasing relation

$$1 + \chi(\mathbf{x}) = F_{\mathbf{x}}[\nabla_i \nabla_j \Phi, \nabla_i \nabla_j \Phi_v; \mathbf{x}],$$

where $\nabla_j \Phi_v \equiv v_j$, one obtains

$$\begin{aligned} 1 + \chi(\mathbf{x}, t) &= \sum_{\mathcal{O}} c_{\mathcal{O}}(t) \mathcal{O}(\mathbf{x}, t) \\ &= c_0 + c_{\delta} + \frac{c_{\delta^2}}{2} s^2 + \frac{c_{s^2}}{2} s^2 + \frac{c_{\delta^3}}{3!} s^3 + \frac{c_{\delta s^2}}{2} s^2 + c_{\psi} s^2 \\ &\quad + \text{stochastic terms} \end{aligned}$$

Remind that stochastic fields ϵ are uncorrelated with all other fields at large s

Bias expansion

$$1 + \delta_X(\boldsymbol{x}, t) = c_0 + c_\delta \delta + \frac{c_{\delta^2}}{2} \delta^2 + \frac{c_{s^2}}{2} s^2 + \frac{c_{\delta^3}}{3!} \delta^3 + \frac{c_{\delta s^2}}{2} \delta s^2 + c_\psi \psi$$

can be written as

$$\delta_X(\boldsymbol{x}) = \sum c_{\mathcal{O}} \mathcal{O}(\boldsymbol{x}),$$

with each operator written in Fourier space at first, second, and third order in

$$\mathcal{O}^{(1)}(\boldsymbol{k}) = K_{\mathcal{O}}^{(1)}(\boldsymbol{k}) \delta^{(1)}(\boldsymbol{k}),$$

$$\mathcal{O}^{(2)}(\boldsymbol{k}) = \int_{12} K_{\mathcal{O}}^{(2)}(\boldsymbol{k}_1, \boldsymbol{k}_2) \delta^{(1)}(\boldsymbol{k}_1) \delta^{(1)}(\boldsymbol{k}_2),$$

$$\mathcal{O}^{(3)}(\boldsymbol{k}) = \int_{123} K_{\mathcal{O}}^{(3)}(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3) \delta^{(1)}(\boldsymbol{k}_1) \delta^{(1)}(\boldsymbol{k}_2) \delta^{(1)}(\boldsymbol{k}_3)$$

Remind that δ is the matter fluctuation and δ_X the tracer fluctuation.

The only operator that appears at first order is δ , hence $\delta_X^{(1)}(\boldsymbol{k}) = c_1 \delta^{(1)}(\boldsymbol{k})$

*We can add a curvature bias ∇^2 , and in such a case $\delta_X^{(1)}(\boldsymbol{k}) = (c_1 - c_\nabla) \delta^{(1)}(\boldsymbol{k}) + c_\nabla \nabla^2 \delta^{(1)}(\boldsymbol{k})$

Operators present at second order in PT

We have the following bias operators

$$\begin{aligned} \text{At least 1st order} & \quad \delta, \\ \text{At least 2nd order} & \quad \delta^2, s^2 \end{aligned}$$

That is, we have $\mathcal{O}^{(2)}(\delta) = \delta^{(2)}, \delta^2(2), s^2(2)$, that we must write as

$$\mathcal{O}^{(2)}(\delta) = \int_{12=} K_{\mathcal{O}}^{(2)}(\mathbf{p}_1, \mathbf{p}_2) \delta^{(1)}(\mathbf{p}_1) \delta^{(1)}(\mathbf{p}_2) = \int \frac{d^3 p}{(2\pi)^3} K_{\mathcal{O}}^{(2)}(\mathbf{p},$$

This is straightforward

$$K_{\delta}^{(2)}(\mathbf{p}_1, \mathbf{p}_2) = F_2(\mathbf{p}_1, \mathbf{p}_2), \quad K_{\delta^2}^{(2)}(\mathbf{p}_1, \mathbf{p}_2) = 1, \quad K_{s^2}^{(2)}(\mathbf{p}_1, \mathbf{p}_2)$$

with

$$S_2(\mathbf{p}_1, \mathbf{p}_2) = \frac{(\mathbf{p}_1 \cdot \mathbf{p}_2)^2}{k_1^2 k_2^2} - \frac{1}{3}.$$

Operators present at third order in PT

We have the following bias operators

$$\begin{aligned} \text{At least 1st order} & \quad \mathcal{O}^{(1)}(x) = x, \\ \text{At least 2nd order} & \quad \mathcal{O}^{(2)}(x) = x^2, s^2, \\ \text{At least 3rd order} & \quad \mathcal{O}^{(3)}(x) = x^3, s^2, \psi, st, s^3 \end{aligned}$$

That is, we have

$$\mathcal{O}^{(3)}(x) = x^3, x^2, s^2, x^3, (s^2)^2, \psi, (st)^2,$$

that we must write as

$$\mathcal{O}^{(3)}(x) = \int_{123=} K_{\mathcal{O}}^{(3)}(x_1, x_2, x_3) \mathcal{O}^{(1)}(x_1) \mathcal{O}^{(1)}(x_2) \mathcal{O}^{(1)}(x_3)$$

Operators present at third order in PT

$$\mathcal{O}^{(3)}(\mathbf{k}) = \int_{123} K^{(3)}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) \delta^{(1)}(\mathbf{k}_1) \delta^{(1)}(\mathbf{k}_2) \delta^{(1)}(\mathbf{k}_3)$$

$$K^{(3)}_1(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = F_3(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3),$$

$$K^{(3)}_2(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = 2F_2(\mathbf{k}_2, \mathbf{k}_3),$$

$$K^{(3)}_3(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = 1,$$

$$K^{(3)}_{s2}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = 2S_2(\mathbf{k}_1, \mathbf{k}_2 + \mathbf{k}_3)F_2(\mathbf{k}_2, \mathbf{k}_3),$$

$$K^{(3)}_{s2}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = S_2(\mathbf{k}_1, \mathbf{k}_2),$$

$$K^{(3)}_{s3}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = S_3(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3),$$

$$K^{(3)}_{st}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = S_2(\mathbf{k}_1, \mathbf{k}_2 + \mathbf{k}_3)(G_2(\mathbf{k}_2, \mathbf{k}_3) - F_2(\mathbf{k}_2,$$

$$K^{(3)}_{\psi}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = G_3(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) - F_3(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3)$$

$$+ 2F_2(\mathbf{k}_2, \mathbf{k}_3) \left(\frac{2}{7} S_2(\mathbf{k}_1, \mathbf{k}_2 + \mathbf{k}_3) - \frac{4}{21} \right)$$

with

$$S_3(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = \left(\frac{k_1^i k_1^j}{k_1^2} - \frac{1}{3} \delta_{ij} \right) \left(\frac{k_2^j k_2^k}{k_2^2} - \frac{1}{3} \delta_{jk} \right) \left(\frac{k_3^k k_3^i}{k_3^2} - \frac{1}{3} \delta_{ki} \right)$$

- Example 1:

$$^{(3)}(\quad) = \int_{123=} F_3(\quad_1, \quad_2, \quad_3) \quad^{(1)}(\quad_1) \quad^{(1)}(\quad_2) \quad^{(1)}(\quad_3)$$

then

$$K_{\delta}^{(3)}(\quad_1, \quad_2, \quad_3) = F_3(\quad_1, \quad_2, \quad_3)$$

.

- Example 2:

$$^{3(3)}(\quad) = \mathcal{FT}[\quad^3(\boldsymbol{x})](\quad) = \int_{123=} \quad^{(1)}(\quad_1) \quad^{(1)}(\quad_2) \quad^{(1)}(\quad_3)$$

then

$$K_{\delta^3}^{(3)}(\quad_1, \quad_2, \quad_3) = 1$$

.

- Example 3:

$$\begin{aligned}
 (s^2 \delta)^{(3)}(\mathbf{x}) &= (s^2 \delta)^{(1)}(\mathbf{x}) = \mathcal{FT}[s^2 \delta^{(1)}(\mathbf{x})](\mathbf{x}) = \int_{\mathbf{p}_2} s^2 \delta^{(1)}(\mathbf{x}) \\
 &= \int \frac{d^3 k_1}{(2\pi)^3} \frac{d^3 k_2}{(2\pi)^3} (2\pi)^3 \delta_D(\mathbf{k}_1 - \mathbf{k}_2) \int_{\mathbf{p}_{12} = \mathbf{x}} S_2(\mathbf{p}_1, \mathbf{p}_2) \\
 &= \int \frac{d^3 k_1 d^3 k_2 d^3 p_1 d^3 p_2}{(2\pi)^6 (2\pi)^6} (2\pi)^3 \delta_D(\mathbf{k}_1 - \mathbf{k}_2) (2\pi)^3 \delta_D(\mathbf{p}_1 - \mathbf{p}_2) \\
 &\quad \times S_2(\mathbf{p}_1, \mathbf{p}_2) \delta^{(1)}(\mathbf{p}_1) \delta^{(1)}(\mathbf{p}_2) \delta^{(1)}(\mathbf{x})
 \end{aligned}$$

Integrating over $\mathbf{p}_1$

$$\begin{aligned}
 (s^2 \delta)^{(3)}(\mathbf{x}) &= \int \frac{d^3 k_2 d^3 p_1 d^3 p_2}{(2\pi)^6} \delta_D(\mathbf{k}_1 - \mathbf{p}_1 - \mathbf{p}_2) S_2(\mathbf{p}_1, \mathbf{p}_2) \delta^{(1)}(\mathbf{p}_1) \\
 &= \int \frac{d^3 k_2 d^3 k_1 d^3 k_3}{(2\pi)^6} \delta_D(\mathbf{k}_1 - \mathbf{k}_3 - \mathbf{k}_2) S_2(\mathbf{k}_1, \mathbf{k}_3) \delta^{(1)}(\mathbf{k}_1) \\
 &= \int \frac{d^3 k_1 d^3 k_2 d^3 k_3}{(2\pi)^9} (2\pi)^3 \delta_D(\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_3) S_2(\mathbf{k}_1, \mathbf{k}_2) \\
 &= \int_{123} S_2(\mathbf{k}_1, \mathbf{k}_2) \delta^{(1)}(\mathbf{k}_1) \delta^{(1)}(\mathbf{k}_2) \delta^{(1)}(\mathbf{k}_3)
 \end{aligned}$$

then

$$K_s^{(3)}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3) = S_2(\mathbf{k}_1, \mathbf{k}_2)$$

matter-tracers cross power spectrum

We construct the cross-spectrum

$$\begin{aligned} (2\pi)^3 \int d^3x \, d^3x' P_{mX}(k) &= \langle X(\mathbf{x}) X(\mathbf{x}') \rangle \\ &= \langle \left(c_\delta + \frac{c_\delta^2}{2} \delta^2 + \frac{c_s^2}{2} s^2 + \frac{c_\delta^3}{3!} \delta^3 + \frac{c_\delta s^2}{2} \delta s^2 + c_\psi \psi + c_{st} s t + \frac{1}{3!} \right) \end{aligned}$$

$$c_\delta \langle \delta(\mathbf{x}) \delta(\mathbf{x}') \rangle' = c_\delta (P_L + P_{22} + P_{13}) = c_\delta P_{1\text{-loop}}^{\text{SPT}}(k)$$

$$\frac{c_{\delta^2}}{2} \langle \varphi^2(\mathbf{x}) \varphi^2(\mathbf{x}') \rangle = \frac{c_{\delta^2}}{2} \left(\langle \varphi^{2(3)}(\mathbf{x}) \varphi^{(1)}(\mathbf{x}') \rangle + \langle \varphi^{2(2)}(\mathbf{x}) \varphi^{(2)}(\mathbf{x}') \rangle \right)$$

First term:

$$\begin{aligned} \langle \varphi^{2(3)}(\mathbf{x}) \varphi^{(1)}(\mathbf{x}') \rangle &= \int_{123=} K_{\delta^2}^{(3)}(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \langle \varphi^{(1)}(\mathbf{x}_1) \varphi^{(1)}(\mathbf{x}_2) \varphi^{(1)}(\mathbf{x}_3) \varphi^{(1)}(\mathbf{x}') \rangle \\ &= \int_{123=} 2F_2(\mathbf{x}_2, \mathbf{x}_3) \langle \varphi^{(1)}(\mathbf{x}_1) \varphi^{(1)}(\mathbf{x}_2) \varphi^{(1)}(\mathbf{x}_3) \varphi^{(1)}(\mathbf{x}') \rangle \end{aligned}$$

$$\begin{aligned} \langle \varphi^{(1)}(\mathbf{x}_1) \varphi^{(1)}(\mathbf{x}_2) \varphi^{(1)}(\mathbf{x}_3) \varphi^{(1)}(\mathbf{x}') \rangle &= (2\pi)^3 \delta(\mathbf{x}_1 + \mathbf{x}_2) \delta(\mathbf{x}_3 + \mathbf{x}') \\ &\quad + (2\pi)^3 \delta(\mathbf{x}_1 + \mathbf{x}_3) \delta(\mathbf{x}_2 + \mathbf{x}') \\ &\quad + (2\pi)^3 \delta(\mathbf{x}_1 + \mathbf{x}') \delta(\mathbf{x}_2 + \mathbf{x}_3) \end{aligned}$$

Inside the integral the first two terms are equal, and the last term is zero since

Performing the integrals over k_2 and k_3 by using the Dirac delta functions, and the dummy variable $k_1 \rightarrow p$, one arrives at

$$\langle P^{(2)}(\hat{k}) P^{(1)}(\hat{k}') \rangle = (2\pi)^3 \delta(\hat{k} - \hat{k}') P_L(k) 4 \int \frac{d^3 p}{(2\pi)^3} F_2(\hat{k}, \hat{p})$$

We develop

$$\int \frac{d^3 p}{(2\pi)^3} F_2(\hat{k}, \hat{p}) P_L(p) = \int \frac{d^3 p}{(2\pi)^3} \left[\frac{17}{21} + \left(x^2 - \frac{1}{3} \right) \frac{x}{2} \left(\frac{p}{k} \right) \right] P_L(p)$$

Second and third terms in the squared brackets vanishes upon angular integration $x = \hat{p} \cdot \hat{k}$ (since they are proportional to the second and third Legendre polynomials)

$$\int \frac{d^3 p}{(2\pi)^3} F_2(\hat{k}, \hat{p}) P_L(p) = \frac{17}{21} \sigma^2$$

and

$$\frac{c_{\delta^2}}{2} \langle P^{(2)}(\hat{k}) P^{(1)}(\hat{k}') \rangle' = \frac{34}{21} c_{\delta^2} \sigma^2 P_L(k)$$

Remind $\sigma^2 = \int \frac{d^3 p}{(2\pi)^3} P_L(k) = \int \frac{d^3 p}{(2\pi)^3} P_L(k) e^{i \cdot 0} = \xi(0)$

The second term in

$$\frac{c_{\delta^2}}{2} \langle \varphi^2(\mathbf{x}) \varphi^2(\mathbf{x}') \rangle = \frac{c_{\delta^2}}{2} \left(\langle \varphi^{2(3)}(\mathbf{x}) \varphi^{(1)}(\mathbf{x}') \rangle + \langle \varphi^{2(2)}(\mathbf{x}) \varphi^{(2)}(\mathbf{x}') \rangle \right)$$

is straightforward.

Second term:

$$\langle \varphi^{2(2)}(\mathbf{x}) \varphi^{(2)}(\mathbf{x}') \rangle' = 2 \int \frac{d^3 p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) P(|\mathbf{p}|) P(p)$$

Hence

$$\frac{c_{\delta^2}}{2} \langle \varphi^2(\mathbf{x}) \varphi^2(\mathbf{x}') \rangle' = c_{\delta^2} \int \frac{d^3 p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) P(|\mathbf{p}|) P(p) + \dots$$

The rest of the terms give

$$\begin{aligned} \frac{1}{2}c_{s^2}\langle s^2(\mathbf{\hat{r}})(\mathbf{\hat{r}}')\rangle' &= c_{s^2}\int\frac{d^3p}{(2\pi)^3}F_2(\mathbf{p},-\mathbf{p})S_2(\mathbf{p},-\mathbf{p})P_L(k)\\ &\quad +2c_{s^2}P_L(k)\int\frac{d^3p}{(2\pi)^3}F_2(\mathbf{p},\mathbf{p})S_2(\mathbf{p},\mathbf{p})\\ \frac{1}{3!}c_{\delta^3}\langle\delta^3(\mathbf{\hat{r}})(\mathbf{\hat{r}}')\rangle' &= \frac{1}{2}c_{\delta^3}\sigma^2P_L(k),\\ \frac{1}{2}c_{\delta s^2}\langle(\delta^2)(\mathbf{\hat{r}})(\mathbf{\hat{r}}')\rangle' &= \frac{1}{3}c_{\delta s^2}\sigma^2P_L(k),\\ c_{st}\langle st(\mathbf{\hat{r}})(\mathbf{\hat{r}}')\rangle' &= 2c_{st}P_L(k)\int\frac{d^3p}{(2\pi)^3}\left[G_2(\mathbf{p},-\mathbf{p})-F_2(\mathbf{p},-\mathbf{p})\right.\\ c_{\psi}\langle\psi(\mathbf{\hat{r}})(\mathbf{\hat{r}}')\rangle' &= 2c_{\psi}P_L(k)\int\frac{d^3p}{(2\pi)^3}\left[\frac{3}{2}G_3(\mathbf{\hat{r}},\mathbf{p},\mathbf{p})-F_3(\mathbf{p},\mathbf{p},\mathbf{p})\right.\\ &\quad \left.-2F_2(\mathbf{p},-\mathbf{p})G_2(\mathbf{p},-\mathbf{p})+F_2(\mathbf{p},\mathbf{p},\mathbf{p})\right]\\ c_{s^3}\langle s^3(\mathbf{\hat{r}})(\mathbf{\hat{r}}')\rangle' &= 0. \end{aligned}$$

matter-tracers cross power spectrum

$$(2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_{mX}(k) = \langle X(\mathbf{k}) X^*(\mathbf{k}') \rangle$$

$$\begin{aligned} P_{mX}(k) = & c_\delta P_L(k) + c_\delta P_{22}(k) + c_\delta P_{13}(k) \\ & + \left(\frac{34}{21} c_{\delta^2} + \frac{1}{2} c_{\delta^3} + \frac{1}{3} c_{\delta s^2} \right) \sigma^2 P_L(k) \\ & + c_{\delta^2} \int \frac{d^3 p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) P(|\mathbf{p}|) P(p) \\ & + c_{s^2} \int \frac{d^3 p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) S_2(\mathbf{p}, -\mathbf{p}) P_L(|\mathbf{p}|) P(p) \\ & + 2c_{s^2} P_L(k) \int \frac{d^3 p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) S_2(\mathbf{p}, -\mathbf{p}) P_L(p) \\ & + 2c_{st} P_L(k) \int \frac{d^3 p}{(2\pi)^3} G_2(\mathbf{p}, -\mathbf{p}) F_2(\mathbf{p}, -\mathbf{p}) S_2(\mathbf{p}, -\mathbf{p}) \\ & + 2c_\psi P_L(k) \int \frac{d^3 p}{(2\pi)^3} \left[\frac{3}{2} G_3(\mathbf{p}, \mathbf{p}, -\mathbf{p}) F_3(\mathbf{p}, \mathbf{p}, -\mathbf{p}) \right. \\ & \quad \left. + 2F_2(\mathbf{p}, -\mathbf{p}) G_2(\mathbf{p}, -\mathbf{p}) F_2(\mathbf{p}, -\mathbf{p}) \right] \end{aligned}$$

Let us define the function

$$\sigma_3^2(k) = \frac{105}{16} \int \frac{d^3p}{(2\pi)^3} \left[S_2(\mathbf{p}, -\mathbf{p}) \left(\frac{2}{7} S_2(-\mathbf{p}, \mathbf{p}) - \frac{4}{21} \right) + \right.$$

Lengthy, but straightforward calculations give

$$c_{s^2} : \quad \int \frac{d^3p}{(2\pi)^3} F_2(-\mathbf{p}, \mathbf{p}) S_2(\mathbf{p}, -\mathbf{p}) P_L(p) = -\frac{8}{21} \sigma_3^2(k) + \frac{34}{63} \sigma_3^2(k)$$

$$c_{st} : \quad \int \frac{d^3p}{(2\pi)^3} \left[G_2(-\mathbf{p}, \mathbf{p}) - F_2(-\mathbf{p}, \mathbf{p}) \right] S_2(\mathbf{p}, -\mathbf{p}) P_L(p) = -\frac{1}{12} \sigma_3^2(k)$$

$$c_\psi : \quad \int \frac{d^3p}{(2\pi)^3} \left[\frac{3}{2} \left[G_3(-\mathbf{p}, \mathbf{p}, \mathbf{p}) - F_3(-\mathbf{p}, \mathbf{p}, \mathbf{p}) \right] \right. \\ \left. - 2F_2(-\mathbf{p}, \mathbf{p}) G_2(\mathbf{p}, -\mathbf{p}) - F_2(\mathbf{p}, -\mathbf{p}) \right] P_L(p) = -\frac{1}{12} \sigma_3^2(k)$$

matter-tracers cross power spectrum

$$\begin{aligned}
 P_{mX}(k) &\ni 2c_{s^2}P_L(k) \int \frac{d^3p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) S_2(\mathbf{p}, -\mathbf{p}) P_L(p) \\
 &+ 2c_{st}P_L(k) \int \frac{d^3p}{(2\pi)^3} G_2(\mathbf{p}, -\mathbf{p}) F_2(\mathbf{p}, -\mathbf{p}) S_2(\mathbf{p}, -\mathbf{p}) P_L(p) \\
 &+ 2c_\psi P_L(k) \int \frac{d^3p}{(2\pi)^3} \left[\frac{3}{2} G_3(\mathbf{p}, \mathbf{p}, -\mathbf{p}) F_3(\mathbf{p}, \mathbf{p}, -\mathbf{p}) \right. \\
 &\quad \left. + 2F_2(\mathbf{p}, -\mathbf{p}) G_2(\mathbf{p}, -\mathbf{p}) F_2(\mathbf{p}, -\mathbf{p}) \right] P_L(p) \\
 &= \left(\frac{16}{21} c_{s^2} + \frac{32}{105} c_{st} + \frac{512}{2205} c_\psi \right) \sigma_3^2(k) P_L(k) \\
 &+ \left(\frac{68}{63} c_{s^2} - \frac{16}{63} c_{st} \right) \sigma^2 P_L(k)
 \end{aligned}$$

matter-tracers cross power spectrum

$$\begin{aligned}
 P_{mX}(k) = & b_{\delta} P_{1\text{-loop}}^{\text{SPT}}(k) \\
 & + b_{\delta^2} \int \frac{d^3 p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) P(|\mathbf{p}|) P(p) \\
 & + b_{s^2} \int \frac{d^3 p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) S_2(\mathbf{p}, -\mathbf{p}) P_L(|\mathbf{p}|) \\
 & + b_{3nl} \sigma_3^2(k) P_L(k)
 \end{aligned}$$

with

$$\begin{aligned}
 b_{\delta} = c_{\delta} + & \left(\frac{34}{21} c_{\delta^2} + \frac{1}{2} c_{\delta^3} + \frac{1}{3} c_{\delta s^2} + \frac{68}{63} c_{s^2} - \frac{16}{63} c_{st} \right) \\
 b_{3nl} = & \frac{16}{21} c_{s^2} + \frac{32}{105} c_{st} + \frac{512}{2205} c_{\psi}
 \end{aligned}$$

and

$$b_{\delta^2} = c_{\delta^2}, \quad b_{s^2} = c_{s^2}.$$

Tracers power spectrum

$$\begin{aligned}
 P(k) = & b_1^2 P_{\text{SPT}}^{1\text{-loop}}(k) + 2b_1 b_2 P_{b_1 b_2}(k) + 2b_1 b_{s2} P_{b_1 b_{s2}}(k) \\
 & + b_2^2 P_{b_2^2}(k) + 2b_2 b_{s2} P_{b_2 b_{s2}}(k) + b_{s2}^2 P_{b_{s2}^2}(k) + 2b_1 b_{3nl} \sigma_3^2 \\
 & k^2 P_L(k),
 \end{aligned}$$

with

$$\begin{aligned}
 P_{b_1 b_2}(k) &= \int \frac{d^3 p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) P_L(\mathbf{p}) P_L(|-\mathbf{p}|), \\
 P_{b_1 b_{s2}}(k) &= \int \frac{d^3 p}{(2\pi)^3} F_2(\mathbf{p}, -\mathbf{p}) S_2(\mathbf{p}, -\mathbf{p}) P_L(\mathbf{p}) P_L(|-\mathbf{p}|), \\
 P_{b_2^2}(k) &= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} P_L(\mathbf{p}) P_L(|-\mathbf{p}|) P_L(p), \\
 P_{b_2 b_{s2}}(k) &= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} P_L(\mathbf{p}) \left[P_L(|-\mathbf{p}|) S(\mathbf{p}, -\mathbf{p}) - \frac{2}{3} P_L(p) \right], \\
 P_{b_{s2}^2}(k) &= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} P_L(\mathbf{p}) \left[P_L(|-\mathbf{p}|) [S(\mathbf{p}, -\mathbf{p})]^2 - \frac{4}{9} P_L(p) \right].
 \end{aligned}$$

Four bias parameters $\{b_1, b_2, b_{s2}, b_{3nl}\}$ and one EFT parameter σ_3 .

Tracers power spectrum

